

TCGA-BRCA Gene Expression Data Retrieval Report

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Date: June 9, 2026

Overview

This report documents the retrieval, preprocessing, and quality control of RNA-sequencing gene expression data from The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) cohort. The data will serve as the primary omics input for downstream weighted gene co-expression network analysis (WGCNA) and a graph neural network based patient similarity framework for breast cancer subtype prediction and survival prognosis modelling.

All data acquisition was performed programmatically via the **TCGAbiolinks** R package. Raw STAR-derived read counts were retrieved in TSV format, processed to retain only protein-coding genes, and split into tumour and normal expression matrices after removal of PAM50 Normal-Like samples. Clinical metadata including receptor status (ER, PR, HER2) and PAM50 molecular subtype labels were extracted and merged into a unified annotation table.

Data Query and Download

Query Parameters

Data were queried from the GDC portal using the following specification:

Parameter	Value
Project	TCGA-BRCA
Data Category	Transcriptome Profiling
Data Type	Gene Expression Quantification
Experimental Strategy	RNA-Seq
Workflow Type	STAR – Counts
Platform	Illumina
Data Format	TSV
Access	Open
Genome Reference	hg38

The query returned **1,231 files** with a total download size of approximately **5.22 GB**. Files were downloaded in chunks of 5 files per archive via the GDC API method to avoid tar extraction failures that occurred at larger chunk sizes on Windows.

Download Notes

The integrity of all files was confirmed by running **GDCprepare** in a **tryCatch** block prior to the main preparation step, which returned no errors.

Gene Annotation and Protein-Coding Filtering

Full Annotation

Gene-level metadata was extracted from the `rowRanges` slot of the `SummarizedExperiment` object returned by `GDCprepare`. The full annotation table contains the following fields for every annotated feature: Ensembl gene ID (`gene_id`), HGNC gene symbol (`gene_name`), and Ensembl biotype (`gene_type`).

Annotation File	Total Features
<code>Gene_Annotation_All.csv</code>	60,660 genes (all biotypes)
<code>Gene_Annotation_ProteinCoding.csv</code>	19,938 genes (protein-coding only)

Protein-Coding Filter

Only genes with `gene_type == "protein_coding"` were retained for downstream analysis. This yielded **19,938 unique protein-coding genes**. Non-coding features excluded include long non-coding RNAs (lncRNAs), pseudogenes, small nuclear RNAs (snRNA), small nucleolar RNAs (snoRNA), microRNAs, and ribosomal RNA genes, which collectively account for the remaining 40,722 annotated features.

Expression Matrix Construction

Raw STAR unstranded read counts were extracted using `assay(ge_data)`, which returns a genes-by-samples integer count matrix. The following preprocessing steps were applied:

1. Rows were subset to the 19,938 protein-coding Ensembl IDs.
2. Sample barcodes (originally 28 characters, e.g. `TCGA-A2-A0CM-01A-11R-A00Z-07`) were trimmed to the standard 16-character vial-level identifier (e.g. `TCGA-A2-A0CM-01A`) using `substr(..., 1, 16)`.
3. Duplicate sample columns arising from barcode trimming were removed by keeping the first occurrence per barcode.
4. Column names were standardised to use hyphens (R internally converts hyphens to periods in some contexts; this was reversed with `gsub("textbackslash\\.", "-", ...)`).
5. Ensembl ID and gene name were prepended as the first two columns so that both identifiers are present throughout all downstream files.

The resulting all-samples matrix (`TCGA-BRCA_raw_counts_protein_coding_all_samples.csv`) has dimensions **19,938 genes × 1,226 samples**.

Clinical Metadata Extraction

Receptor Status

Hormone and growth factor receptor status was obtained from the BCR Biotab clinical supplement (`clinical_patient_brca` table), downloaded via a separate `GDCquery` call with

`data.format = "BCR Biotab"`. The first two rows of this table are annotation header artefacts and were discarded. The extracted fields were `er_status_by_ihc`, `pr_status_by_ihc`, and `her2_status_by_ihc`, all measured by immunohistochemistry (IHC). The distribution of receptor status labels across the cohort is shown below.

Status	ER	PR	HER2
Positive	899	779	187
Negative	263	380	627
Equivocal / Indeterminate	3	5	205
Not Evaluated / Not Available	60	61	206

PAM50 Molecular Subtype and Sample Type

PAM50 subtype labels and sample type annotations were retrieved using `colDataPrepare(sample_ids)`, which queries the GDC API for clinical and molecular paper annotations associated with each sample barcode.

PAM50 Subtype	Sample Count
Luminal A	567
Luminal B	209
Basal-like	195
HER2-enriched	82
Normal-Like	40
Not Available (NA)	132
Total	1,225

Sample Type	Count
Primary Tumor	1,105
Solid Tissue Normal	113
Metastatic	7
Total	1,225

Both tables were merged on the `patient` barcode field, producing a unified clinical meta-data table (`TCGA-BRCA_clinical_metadata.csv`) with 1,225 records and 7 fields: `patient`, `ER_Status`, `PR_Status`, `HER2_Status`, `sample`, `sample_type`, and `paper_BRCA_Subtype_PAM50`.

Normal-Like Sample Removal and Final Split

PAM50 Normal-Like tumour samples ($n = 40$) were identified and removed from the expression matrix prior to the tumour/normal split. These samples cluster transcriptionally with normal breast tissue rather than with any of the four canonical tumour subtypes, and their inclusion would confound subtype classifiers and co-expression module detection. Removal was performed at the sample barcode level using the `paper_BRCA_Subtype_PAM50 == "Normal"` criterion from the merged clinical table.

The 7 metastatic samples were excluded implicitly, as they do not appear in the `Primary Tumor` category used to define the tumour set.

The final expression matrices were exported with Ensembl ID and gene name as the leading two columns, followed by one column per sample.

Output File	Sample Type	Genes	Samples
TCGA-BRCA_protein_coding_tumor.csv	Primary Tumor (PAM50 filtered)	19,938	1,065
TCGA-BRCA_protein_coding_normal.csv	Solid Tissue Normal	19,938	113

Both matrices passed the final quality control check: no missing values (NA) were detected in either matrix.

Summary of Output Files

File	Description
Gene_Annotation_All.csv	All 60,660 annotated features with Ensembl ID, gene symbol, and bio-type
Gene_Annotation_ProteinCoding.csv	Filtered set of 19,938 protein-coding genes
TCGA-BRCA_clinical_metadata.csv	Merged receptor status, PAM50 subtype, and sample type for 1,225 samples
TCGA-BRCA_protein_coding_tumor.csv	Final tumour expression matrix (1,065 samples, 19,938 genes)
TCGA-BRCA_protein_coding_normal.csv	Final normal expression matrix (113 samples, 19,938 genes)